

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	35% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	AB Dhanyashree	Reg. No.:	192472002
Section / Batch:	D/2024	Date:	11.08.26

Code of Conduct

I AB Dhanyashree(192472002) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging, HMM	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q1

Annexure A – SCENARIO BASED

A company is developing an NLP-based grammar checker for educational applications. The system is trained using a manually POS-tagged corpus. Your task is to recreate the Hidden Markov Model (HMM) **without using any built-in NLP or HMM libraries**.

From the given corpus,

- determine the **Emission Probability**
- determine the **Transition Probability**
- implement the **Viterbi Algorithm**
- predict the POS tags for the new input sentence.

Use only Python dictionaries, loops, and basic programming constructs.

Training Corpus

Sentence	POS Tags
The/DT boy/NN eats/VBZ rice/NN	DT NN VBZ NN
The/DT girl/NN drinks/VBZ milk/NN	DT NN VBZ NN
A/DT cat/NN drinks/VBZ milk/NN	DT NN VBZ NN
The/DT dog/NN chases/VBZ cat/NN	DT NN VBZ NN
A/DT teacher/NN teaches/VBZ students/NNS	DT NN VBZ NNS
Students/NNS study/VBP English/NN	NNS VBP NN
Birds/NNS fly/VBP high/RB	NNS VBP RB
Children/NNS play/VBP games/NNS	NNS VBP NNS

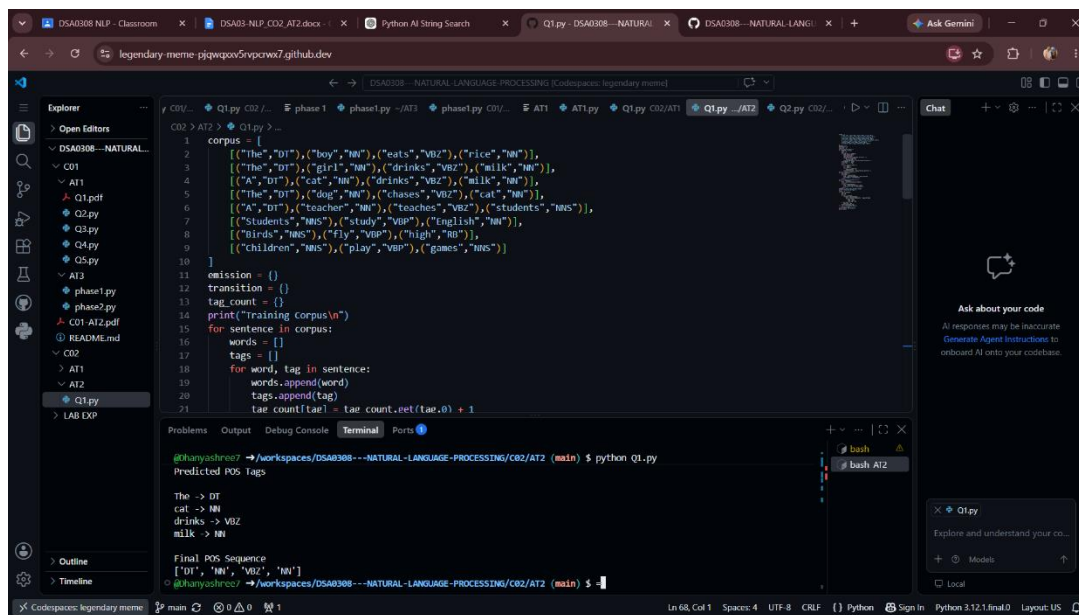
New Input Sentence

The cat drinks milk

Tasks

1. Separate each sentence into words and POS tags.
2. Compute the **Emission Probability** of every word.
3. Compute the **Transition Probability** between POS tags.
4. Recreate the Hidden Markov Model using Python dictionaries.
5. Implement the **Viterbi Algorithm** without using any built-in HMM/NLP libraries.
6. Predict the POS tag sequence for “The cat drinks milk”

OUTPUT:



```
1 corpus = [
2     [{"The", "DT"}, {"boy", "NN"}, {"eats", "VBZ"}, {"rice", "NN"}],
3     [{"The", "DT"}, {"girl", "NN"}, {"drinks", "VBZ"}, {"milk", "NN"}],
4     [{"A", "DT"}, {"cat", "NN"}, {"drinks", "VBZ"}, {"milk", "NN"}],
5     [{"The", "DT"}, {"dog", "NN"}, {"chases", "VBZ"}, {"cat", "NN"}],
6     [{"A", "DT"}, {"teacher", "NN"}, {"teaches", "VBZ"}, {"students", "NNS"}],
7     [{"Students", "NNS"}, {"study", "VBP"}, {"English", "NN"}],
8     [{"Birds", "NNS"}, {"fly", "VBP"}, {"high", "RB"}],
9     [{"Children", "NNS"}, {"play", "VBP"}, {"games", "NNS"}]
10 ]
11 emission = {}
12 transition = {}
13 tag_count = {}
14 print("Training corpus\n")
15 for sentence in corpus:
16     words = []
17     tags = []
18     for word, tag in sentence:
19         words.append(word)
20         tags.append(tag)
21     tag_count[tag] = tag_count.get(tag, 0) + 1
```

```
@hanyashree7 ~/workspaces/DSA0308---NATURAL-LANGUAGE-PROCESSING/C02/AT2 (main) $ python Q1.py
Predicted POS tags

The -> DT
cat -> NN
drinks -> VBZ
milk -> NN

Final POS Sequence
['DT', 'NN', 'VBZ', 'NN']
```

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1	15	20	25	20	15	95	95

Faculty In-charge	Course Coordinator	Program Director
Kumaragurubaran T		
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____